EXPERIMENT – 3

Flowchart for the Operation of a Network Switching Subsystem (NSS)

AIM

To develop a flowchart in Figma illustrating the operation of the Network Switching Subsystem (NSS) in a mobile cellular system, including subscriber authentication, call routing, connection establishment, call termination, and handling of unsuccessful calls.

PROCEDURE

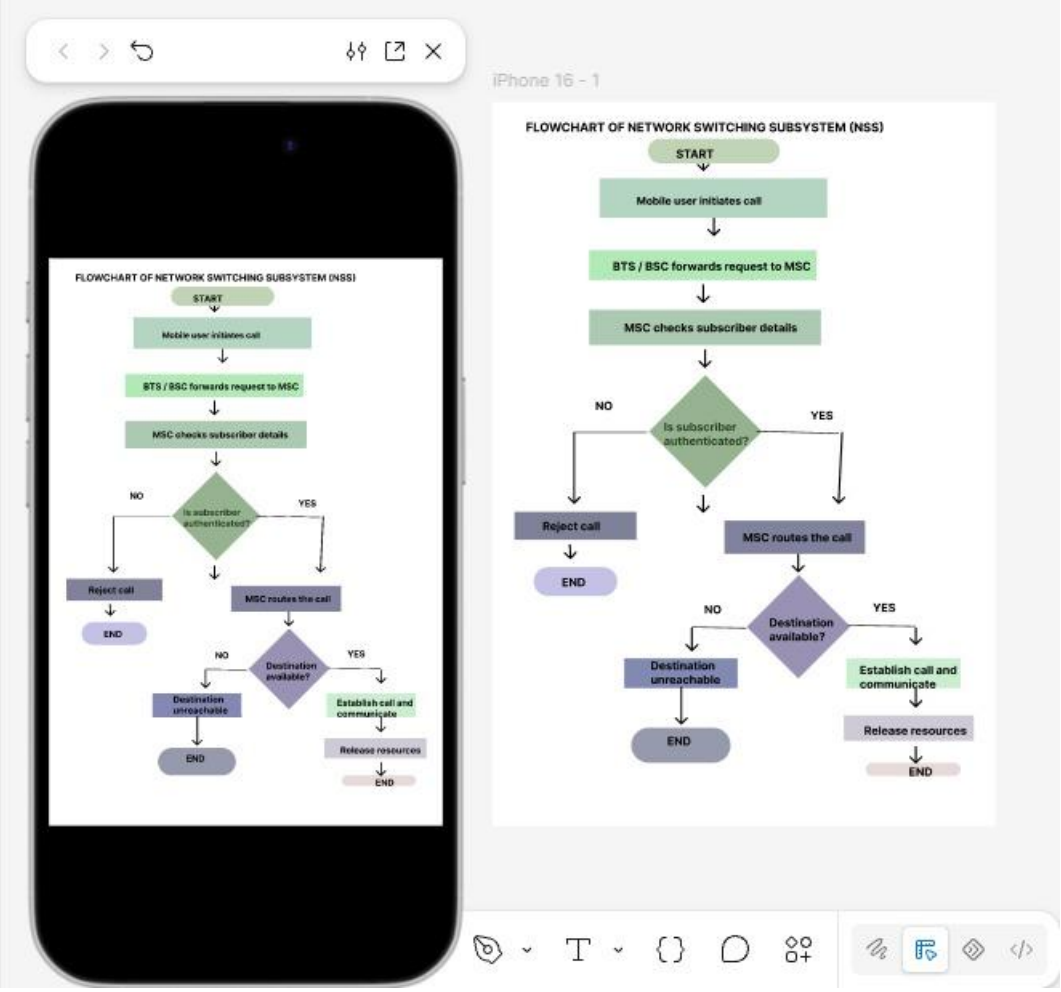
1. Open Figma (figma.com) and create a new design file.
2. Add the title “FLOWCHART OF NETWORK SWITCHING SUBSYSTEM (NSS)” at the top of thecanvas.
3. Create the Start and End symbols, process rectangles, and decision symbols. Use rectangles foroperations and diamonds for decision points.
4. Add the process labels in sequence: Start; Mobile user initiates call; BTS/BSC forwards therequest to MSC; MSC checks subscriber details; Is subscriber authenticated?; MSC routes the call; Destination available?; Establish call and communicate; Release resources; End.
5. Connect each step using directional arrows. From “Is subscriber authenticated?”, connect YES tocall routing and NO to Reject call, followed by End.
6. From “Destination available?”, connect YES to connection establishment and NO to Destinationunreachable, followed by End.
7. Use consistent colours to distinguish start/end, processes, and decisions. Align the shapes andkeep the arrows clear.
8. Review the flowchart for correct sequence and labels. Save the Figma file and export the design asPNG or PDF if required.

DESIGN

The flowchart follows a top-to-bottom structure. A mobile-originated call begins at START and proceeds through the radio access network (BTS/BSC) to the MSC. The MSC checks subscriber information with the relevant subscriber databases and authentication functions. If authentication fails, the call is rejected. If authentication succeeds, the MSC routes the call and checks whether the destination is available. An available destination leads to call connection and communication; otherwise, the call is marked unreachable. When the conversation ends, network resources are released and the process terminates.

Design symbols: Oval/rounded capsule – Start or End; Rectangle – process/action; Diamond – decision; Arrow – direction of control flow. Use YES and NO labels on decision branches.

FLOWCHART



RESULT

The Network Switching Subsystem (NSS) operation flowchart was designed in Figma. It illustrates the call-processing sequence, subscriber authentication, call routing, successful connection, and the paths followed when authentication or destination availability fails.